

Analogies — Chemistry Chapter 6: Stoichiometry

Every abstract idea in this chapter, reframed as something you already understand. When a concept feels slippery, come here — these are your mental handles.

Topic 1 — Empirical vs Molecular Formula

The cookie recipe

Imagine Grandma's chocolate chip cookie recipe:

- **Full recipe (molecular):** 2 cups flour, 4 eggs, 6 tablespoons sugar, makes 24 cookies
- **Simplified recipe (empirical):** 1 part flour, 2 parts eggs, 3 parts sugar

Both describe the same cookies, just at different levels of detail. The empirical formula is Grandma saying "for every 1 cup of flour, use 2 eggs and 3 tablespoons of sugar." The molecular formula is Grandma saying "here's exactly how much to make 24 cookies."

Neither is "right" or "wrong" — they're answering different questions.

The school photo

Your class photo shows 30 students. You could describe the photo as:

- **Molecular:** "There are 18 girls and 12 boys in the photo."
- **Empirical:** "The girl-to-boy ratio is 3:2."

The empirical version is great if you're comparing classes — but it doesn't tell you how big each class is. The molecular version is exact — but you'd have to do extra work to compare to another class.

Chemistry uses both for exactly the same reasons.

The music chord

A major chord in music has a specific ratio: root, third, fifth. A C major chord is C-E-G. A G major chord is G-B-D.

- **Empirical:** "1 root : 1 third : 1 fifth" — the universal recipe for a major chord.
- **Molecular:** "C-E-G" or "G-B-D" — the specific notes in one chord.

Different keys, same structural ratio. Empirical = the pattern. Molecular = this specific instance of the pattern.

Topic 2 — Molecular Mass & Formula Mass

Weighing a LEGO creature

You can't weigh one LEGO brick by looking at a picture of it. But you CAN calculate the weight of a LEGO dragon if you know:

1. Which bricks are in it.
2. How much each brick weighs.

So if a dragon has 2 red bricks (1 gram each), 5 yellow bricks (2 grams each), and 1 green brick (3 grams), the dragon weighs: $2(1) + 5(2) + 1(3) = 15$ grams.

That's exactly how molecular mass works. You count up each "brick" (atom) in the "creature" (molecule), multiply by its weight (atomic mass), and add them up.

The grocery receipt

A molecular mass calculation is just a grocery receipt:

Item	Qty	Price	Total
Hydrogen	2	\$1	\$2
Oxygen	1	\$16	\$16
Total			\$18

Congratulations — you just calculated the molecular mass of water ($\text{H}_2\text{O} = 18$ amu). The atoms are items, the atomic masses are prices, and molecular mass is the total.

The weighing scale for packages

Imagine you're shipping a box of books. You can't weigh individual pages, but if you know: - The book has 200 pages - Each page weighs 5 grams - The cover weighs 100 grams

You can calculate: $(200 \times 5) + 100 = 1100$ grams. That's the weight of one book.

A molecule is like that book. You count its "pages" (atoms) and "cover" (bonds, which we ignore in mass calculations because they're massless) and tally up the weight.

Topic 3 — The Mole & Avogadro's Number

The dozen, upgraded

We already use counting units every day: - **1 pair** = 2 (you buy shoes in pairs) - **1 dozen** = 12 (you buy eggs in dozens) - **1 gross** = 144 (pencils come in gross) - **1 ream** = 500 (paper comes in reams) - **1 mole** = 6.022×10^{23} (atoms come in moles)

Each counting unit is designed to match the size of what you're counting. Eggs come in dozens because 12 is convenient. Paper comes in reams because you buy it 500 sheets at a time. Atoms come in moles because 6.022×10^{23} atoms finally add up to something you can hold in your hand.

The mole isn't weird. It's just a really, really big counting unit, designed for really, really small things.

The popcorn jar

Imagine a gigantic jar of popcorn. If I asked you to count every kernel, you'd say "no way, there are too many." But if I asked you to weigh the jar, that's easy.

You know 1 kernel weighs about 0.1 gram. The jar weighs 600 grams. So there are $600 \div 0.1 = \mathbf{6000}$ kernels in the jar.

You just counted 6000 things without counting a single one. You used **weight** to find the **count**.

That's exactly what chemists do with moles. They can't count atoms, but they can weigh them. And since 1 mole of carbon weighs exactly 12 grams, if you have 24 grams of carbon, you have 2 moles = $2 \times 6.022 \times 10^{23}$ atoms. Counting by weighing.

The pennies in a bag

Here's another version. A single penny weighs 2.5 grams. You have a bag of pennies that weighs 1000 grams. How many pennies?

$1000 \div 2.5 = 400$ pennies.

Easy, right? No counting, just a scale.

Now scale it up to atoms. You have 24 grams of carbon. One mole of carbon weighs 12 grams. So: $24 \div 12 = \mathbf{2}$ moles of **carbon**. Which is $2 \times 6.022 \times 10^{23} = 1.2 \times 10^{24}$ atoms. No counting, just a scale.

It's the same trick. The bigger the thing you're counting, the bigger the counting unit. Pennies → just use numbers. Eggs → use dozens. Atoms → use moles.

Avogadro's number is cosmically big

To give you a sense of scale:

- If you had one mole of grains of sand, it would cover the entire United States **30 meters deep**.
- If you had one mole of pennies and split them evenly among everyone on Earth, each person would get enough pennies to make them a **billion times richer than the richest billionaire on Earth today**.
- If you stacked one mole of sheets of paper, the stack would stretch from Earth to the nearest star (Proxima Centauri) **and back, many times**.

That's just how ridiculously huge Avogadro's number is. And yet, in chemistry, one mole of water fits in a tablespoon. That's how tiny atoms are.

Topic 4 — Molar Mass: The Bridge

The currency converter

Imagine you're traveling from the land of Grams to the land of Moles. At the border, there's a currency exchange booth. The exchange rate depends on what substance you're carrying:

- **If you're carrying water:** 18 g = 1 mole (exchange rate: 18)

- **If you're carrying sodium:** 23 g = 1 mole (exchange rate: 23)
- **If you're carrying CO₂:** 44 g = 1 mole (exchange rate: 44)

The exchange rate IS the molar mass. If you know the exchange rate, you can always convert between the two currencies.

- 36 grams of water → $36 \div 18 = \mathbf{2\ moles}$ of water
- 5 moles of sodium → $5 \times 23 = \mathbf{115\ grams}$ of sodium

You never need to memorize anything fancy. Just the exchange rate for each substance — which you calculate from its molecular mass.

The bag of flour

A grocery store sells flour in 1-kilogram bags. You need 3 kg. How many bags? **3 bags**.

You need 500 grams. How many bags? **0.5 bag**.

Now swap "bag" for "mole" and "kg" for "grams of X." If you need 36 grams of water, and 1 mole weighs 18 grams, you need $36 \div 18 = 2$ "bags" (moles) of water.

The molar mass is just the "size of the bag" for each substance.

The elevator capacity sign

An elevator has a sign: "Maximum capacity: 10 people or 800 kg."

Why both? Because sometimes you know the number of people, and sometimes you only know the total weight. As long as you know the average weight per person (80 kg), you can convert between them.

In chemistry: - "Number of people" = number of moles - "Total weight" = mass in grams - "Weight per person" = molar mass
 And you convert between them using `n = mass ÷ molar mass` , just like `people = weight ÷ weight per person` .

Topic 5 — Writing Ionic Formulas

The dance floor rule

Picture a middle-school dance. There are kids with "+2" shirts (want to dance with 2 people) and kids with "-1" shirts (want 1 partner). The rule: everyone must be paired up, no one left out, total + charges = total − charges.

A "+2" kid needs TWO "-1" kids to dance with. So for every 1 Mg²⁺, you need 2 Cl[−]. Formula: **MgCl₂**.

A "+3" kid needs THREE "-1" kids. So for every 1 Al³⁺, you need 3 Cl[−]. Formula: **AlCl₃**.

What about two "+3" kids and three "-2" kids? $2 \times (+3) = +6$, $3 \times (−2) = −6$, balanced. Formula: **Al₂O₃**.

Every ionic formula is just a dance-floor pairing where the total charges cancel out.

The seesaw

Imagine a seesaw. On one side, positive ions. On the other side, negative ions. Each ion has a "weight" equal to its charge.

- Na⁺ (weight 1) + Cl[−] (weight 1) → balanced, 1:1
- Mg²⁺ (weight 2) + 2 Cl[−] (weight 1 each, total 2) → balanced, 1:2
- Al³⁺ (weight 3) + 3 Cl[−] (total 3) → balanced, 1:3
- 2 Al³⁺ (total 6) + 3 O^{2−} (total 6) → balanced, 2:3

You keep adding ions to each side until the seesaw is level. That's your formula.

The building puzzle

Think of positive ions as "plus blocks" (+1, +2, +3) and negative ions as "minus blocks" (−1, −2, −3). You need to build a structure where total plus = total minus.

For Al³⁺ and O^{2−}: imagine the LCM of 3 and 2 is 6. So you need 6 units of + and 6 units of −. That's $2 \times (+3) = +6$ and $3 \times (−2) = −6$. → Al₂O₃.

You're essentially solving "find the lowest common multiple" every time you write an ionic formula.

Topic 6 — Balancing Chemical Equations

The seesaw (again, but different)

A chemical equation is a seesaw. The reactants are on one side, the products on the other. Each element has to have the same

count on both sides, or the seesaw tips.

Imagine: on the left, 2 hydrogen atoms and 1 oxygen atom. On the right, 1 hydrogen atom and 2 oxygen atoms. That's unbalanced — the seesaw is crashing.

Your job: add coefficients (like adding extra weight blocks) until both sides have exactly the same number of each element.

The Lego rebuild

Imagine you have a Lego house made of 10 red bricks, 4 blue bricks, and 2 yellow bricks. You take it apart and rebuild it into something new — say, a Lego car.

Here's the key: **you must use every single brick**. You can rearrange them, but not throw any away or add new ones.

So the car must also have 10 red, 4 blue, 2 yellow. Any other combination is impossible — the bricks are conserved.

That's exactly what balancing chemical equations does. Atoms are Lego bricks. Reactions are rebuilding projects. You can rearrange, but you can't vanish or add atoms out of thin air.

The kitchen inventory

Pretend you're running a small restaurant. Every morning, you write down exactly how many tomatoes, onions, and potatoes are in your fridge. Every evening, you count again. If the count doesn't match (minus what you used in cooking), something's wrong — either someone stole food or your math is off.

Reactions are the same. You start with certain atoms (morning inventory) and end with those same atoms in new combinations (evening inventory, plus "used in cooking"). If the atoms don't match, your equation is wrong.

Coefficients are multipliers, subscripts are identity

Here's a crucial analogy.

Imagine ordering pizza for a party. You call the pizzeria and say "2 large Hawaiian pizzas."

- **Coefficient = 2** (you want 2 pizzas)
- **Subscript-like info** = "Hawaiian" (the type of pizza)

If you say "3 large Hawaiian" instead, you get 3 pizzas of the same type. Changing the coefficient just means more pizzas.

But if you change "Hawaiian" to "Pepperoni," you've ordered a totally different pizza. The identity of what you're ordering has changed.

In chemical formulas: - Coefficient (2 H₂O) means "2 copies of water" — more water. - Subscript change (H₂O → H₂O₂) means "different compound" — peroxide instead of water.

Same rule as the pizza: coefficient = how many, subscript = what.

Topic 7 — Ionic Equations

The soccer game with referees and fans

Picture a soccer game. On the field, there's a group of players actually playing — running, kicking, scoring. In the stands, there are fans cheering. The fans are important for atmosphere, but they're not part of the game. If you wanted to describe "what happened in the game," you'd focus on the players, not the fans.

That's what net ionic equations do. The **players** are the ions actually reacting. The **fans** are the spectator ions — they're there, but they don't participate. A net ionic equation cuts out the fans and shows you only the players.

The movie with only the main characters

Imagine a 3-hour superhero movie, but you only want the highlights — the main battle, the big reveals. If you edit out every scene that doesn't involve the main characters, you'd end up with a 10-minute highlight reel that captures the whole story.

Net ionic equations are highlight reels. They cut out everything that doesn't matter and show you ONLY what's actually happening chemically.

The group project

You're working on a group project with 5 people. Two of them actually do the work; three of them just showed up to put their names on the paper. If you had to describe "who did this project," you'd mention the two workers and ignore the three spectators.

In **HCl + NaOH → NaCl + H₂O** : - **Workers:** H⁺ and OH⁻ (they combine to form water) - **Spectators:** Na⁺ and Cl⁻ (they're there but don't do anything)

Net ionic equation: $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$. Credit where credit's due.

Topic 8 — Molecular vs Structural Formula

The blueprint vs the inventory

Imagine you're building a house. You need two kinds of documents:

- **The inventory list:** 1000 bricks, 200 planks, 50 windows, 3 doors. This tells you how many of each item you need.
- **The blueprint:** Shows how the bricks, planks, windows, and doors are ARRANGED to form a house.

If you only have the inventory, you might use the same materials to build a very different house — or a shed, or even a ramp. The inventory doesn't guarantee a specific shape.

That's exactly like the difference between molecular and structural formulas. The **molecular formula** is the inventory (counts of each atom). The **structural formula** is the blueprint (how the atoms are arranged).

Twins vs identical twins

Fraternal twins share a birthday and parents, but they look different. Identical twins share a birthday, parents, AND DNA, and look the same.

In chemistry, compounds with the same molecular formula but different structures are called **isomers** — they're like fraternal twins. Same "genetics" (same atoms, same counts), but different appearance (different arrangements).

Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) and dimethyl ether (CH_3OCH_3) are fraternal-twin isomers: both $\text{C}_2\text{H}_6\text{O}$, but one is a drink and the other is a gas.

The anagram

Take the letters E, A, T, S. You can arrange them to spell: - EATS - SATE - SEAT - TEAS

Same letters, different arrangements, different words. The "molecular formula" is the letter count: 1 E, 1 A, 1 T, 1 S. But the "structural formula" is the specific word you spelled.

Compounds work the same way — same atoms can "spell" different molecules.

The meta-analogy: why stoichiometry exists at all

Here's the biggest analogy of all. Imagine you're trying to run a restaurant blindfolded. You know recipes ("2 eggs + 1 cup flour = pancake"), but you can't see your ingredients. You can only weigh things.

How would you make pancakes?

You'd figure out: "1 egg weighs 50 g, 1 cup of flour weighs 120 g. So 100 g of egg + 120 g of flour = 1 pancake batch." You'd invent a system to convert weights into "recipe units."

That's exactly what chemistry had to do. Atoms are invisible. Reactions follow specific recipes (like " $2\text{H}_2 + 1\text{O}_2 = 2\text{H}_2\text{O}$ "). But we can only weigh stuff. So chemists invented the mole to bridge weight → recipe units. It's the blindfolded chef's workaround.

The whole chapter — empirical formulas, molecular masses, molar masses, balanced equations — is one giant system for cooking chemistry by weight, in a world where you can't see your ingredients.

That's why it's beautiful. Stoichiometry is the art of cooking with invisible atoms.